

Local Search: Local Beam Search and Genetic Algorithms Part-7

Introduction

In the realm of artificial intelligence, finding optimal solutions to complex problems often involves navigating incredibly vast search spaces. Traditional systematic search algorithms, while guaranteeing optimality, frequently fail due to prohibitive memory and time requirements when faced with large-scale problems. This challenge necessitates the adoption of more pragmatic approaches, such as local search algorithms. These methods, including hill climbing, random walk, and simulated annealing, offer asymptotic optimality, meaning they can find the global optimum given infinite time. However, their basic forms often operate with a single "current node," limiting their exploratory power. This lecture delves into advanced local search techniques—Local Beam Search and Genetic Algorithms—that enhance exploration and robustness by managing multiple candidate solutions simultaneously, drawing inspiration from natural selection to tackle some of the most challenging optimization tasks.

Core Concepts

Term	Definition	Significance
Local Beam Search	An extension of hill climbing that maintains a "beam" of K best states at each step, exploring their successors and selecting the K best from the combined set to form the next beam.	Addresses the memory constraint of systematic search by keeping a limited number of promising states, improving exploration over single-node hill climbing.
Stochastic Local Beam Search	A variant of local beam search where the K successors for the next beam are chosen probabilistically, biased towards better-performing states, rather than strictly selecting the K absolute best.	Mitigates the loss of diversity inherent in standard local beam search, preventing the algorithm from converging too quickly on a single local optimum.

Term	Definition	Significance
Genetic Algorithm (GA)	An optimization algorithm inspired by natural selection, where a "population" of candidate solutions evolves over generations through processes like selection, crossing over, and mutation.	Offers a powerful, non-local search mechanism by combining features of multiple "parent" solutions, enabling large jumps in the search space and potentially escaping local optima.
Population	In genetic algorithms, the set of K candidate solutions (states) maintained at any given time, analogous to a generation of individuals in biological evolution.	Represents the diversity of solutions being explored, with fitter individuals having a higher chance of contributing to the next generation.
Fitness Function	A function that evaluates the "goodness" or quality of each candidate solution (state) in the population, guiding the selection process.	Quantifies the performance of a solution, allowing the algorithm to differentiate between better and worse states for selection and reproduction.
Crossing Over	A genetic operator where genetic material (parts of the string representation) from two "parent" solutions is exchanged to create new "child" solutions.	Enables the combination of desirable traits from different solutions, facilitating non-local exploration and the discovery of novel, potentially superior, configurations.
Mutation	A genetic operator that introduces small, random changes to a solution's string representation, analogous to random genetic mutations in biology.	Maintains diversity in the population and helps prevent premature convergence by introducing new features that might not be present in the current gene pool.
Champion	The best-performing state found so far in a genetic	Ensures that the best solution discovered is preserved,

Term	Definition	Significance
	algorithm's population, which is often explicitly carried forward to the next generation to prevent its loss.	preventing regression in the search process and guaranteeing monotonic improvement in the best-found solution.
Mixed Ability	A theoretical concept suggesting that organisms (or algorithms) might optimize for the ability to reproduce successfully with a wide range of partners, rather than solely for individual fitness.	Provides an alternative perspective on evolutionary success, potentially explaining why genetic algorithms, which strictly optimize for fitness, might not always outperform simpler methods.
Gradient Descent	An iterative optimization algorithm used to find the local minimum of a function by taking steps proportional to the negative of the gradient (or positive for ascent) of the function at the current point.	Represents the continuous space analogue of local search, applying the same core intuition of moving towards better solutions in a continuous landscape.

Detailed Analysis

1. From Single-Node to Beam Search

Traditional local search algorithms like hill climbing operate with a single current node, making local changes to find a better state. This approach, while memory-efficient, can easily get stuck in local optima. To overcome this, the concept of maintaining multiple candidate solutions simultaneously emerged, leading to Local Beam Search.

- **Local Beam Search (LBS):**
- Instead of one current node, LBS maintains a "beam" of **K** states.
- Initially, **K** states are randomly selected.
- In each iteration, all successors for all **K** states in the current beam are generated.

- From this combined pool of successors, the K best states are selected to form the new beam for the next iteration.
- This process continues until a goal state is found or a termination condition is met.
- **Limitation:** A critical issue with LBS is the potential loss of diversity. If multiple states in the beam are on the same "hill," their successors might also be on that same hill, leading to the entire beam converging to a single local optimum. This makes it behave similarly to K parallel hill-climbing searches that are not truly independent.
- **Stochastic Local Beam Search:**
 - To address the diversity problem, stochastic LBS introduces probabilistic selection.
 - Instead of strictly picking the K best successors, it selects K successors randomly, but with a bias towards better-performing ones. This means a slightly worse successor might still be chosen, maintaining diversity.
 - This approach is analogous to natural selection, where fitter individuals have a higher probability of survival and reproduction, but not a guaranteed one.

2. Genetic Algorithms: Mimicking Natural Evolution

Genetic Algorithms (GAs) take the inspiration from natural selection a step further, introducing mechanisms for combining solutions rather than just selecting from successors. GAs are a form of stochastic local beam search where successor generation involves combining two parent states.

- **State Representation:**
 - In GAs, a state is typically represented as a "string" over a finite alphabet, often called a chromosome or genotype.
 - For example, in the 8-Queens problem, a state can be represented as a string of 8 numbers, where each number indicates the row position of a queen in a specific column (e.g., [1, 6, 2, 5, 7, 4, 8, 3] for an 8-Queens board).
- **Population and Fitness Function:**
 - GAs start with an initial "population" of K randomly generated states (strings).
 - A "fitness function" evaluates the quality of each state. For the 8-Queens problem, this could be the number of non-attacking pairs of queens. Higher fitness values indicate better solutions.

- **Evolutionary Steps:** GAs simulate evolution through three main steps:
- **Selection:** Parents are chosen from the current population for reproduction. This selection is probabilistic, with fitter individuals having a higher chance of being selected. For instance, if a state has 31% fitness, it has a 31% chance of being selected as a parent. This ensures "natural selection" where less fit parents are less likely to contribute to the next generation.
- **Crossing Over (Recombination):** Two selected parent strings are combined to produce one or more "child" strings. A "crossing over point" is randomly chosen, and segments of the parent strings are swapped.
- Example: Parents $P1 = [3, 2, 7, 4, 8, 5, 5, 2]$ and $P2 = [2, 4, 7, 5, 2, 4, 1, 1]$
- If crossing over point is 3:
- Child 1: $[3, 2, 7]$ (from P1) + $[5, 2, 4, 1, 1]$ (from P2) = $[3, 2, 7, 5, 2, 4, 1, 1]$
- Child 2: $[2, 4, 7]$ (from P2) + $[4, 8, 5, 5, 2]$ (from P1) = $[2, 4, 7, 4, 8, 5, 5, 2]$
- Crossing over allows for "non-local jumps" in the search space, combining good features from different solutions.
- **Mutation:** After crossing over, small, random changes are introduced into the child strings with a low probability.
- Example: In a child string $[3, 2, 7, 5, 2, 4, 1, 1]$, a mutation might change the 5 at position 4 to a 1, resulting in $[3, 2, 7, 1, 2, 4, 1, 1]$.
- Mutation helps maintain diversity and prevents the algorithm from getting stuck in local optima by introducing novel genetic material.
- **Champions:** To prevent the loss of the best solution found so far, GAs often employ an "elitism" strategy, where the "champion" (the fittest individual) from the current population is directly carried over to the next generation.
- **Critique of Genetic Algorithms:**
- **Positive:** GAs excel at exploration due to crossing over, which allows for significant, non-local jumps in the search space by combining good partial solutions. This can help escape local optima that simpler local search methods might get stuck in.
- **Negative:**
- **Tunable Parameters:** GAs have many parameters (population size K , mutation probability, crossing over probability, number of parents, etc.) that

are difficult to tune, and the algorithm's performance can be highly sensitive to these choices.

- **Empirical Evidence:** Historically, there was a lack of convincing evidence that GAs consistently outperformed simpler methods like hill climbing with random restarts across a wide range of problems.
- **Historical Context:** The field of AI has seen shifts in popularity. GAs were prominent in the 80s and 90s but faced ostracization from mainstream machine learning conferences due to perceived lack of formality and reproducibility. However, recent successes in areas like training neural networks and reinforcement learning agents have led to a resurgence of interest.
- **"Mixed Ability" Theory:** Some research suggests that biological evolution might not solely optimize for individual fitness but also for "mixed ability"—the capacity to reproduce successfully with diverse partners. This contrasts with GAs' strict fitness optimization and could be a reason for their historical limitations.

3. Local Search in Continuous Space: Gradient Descent

The fundamental intuitions of local search extend beyond discrete problems into continuous optimization. When dealing with continuous variables, such as optimizing parameters in a neural network, the concept of "local moves" translates to moving along the gradient of a function.

- **Gradient Descent/Ascent:**
 - Instead of discrete neighbors, the "direction of improvement" in continuous space is given by the gradient of the objective function.
 - **Gradient Descent** moves in the direction opposite to the gradient to find a minimum, while **Gradient Ascent** moves in the direction of the gradient to find a maximum.
 - The "step size" or "learning rate" (often denoted as λ) determines how far to move in the gradient's direction.
 - **Line Search:** Algorithms like line search are used to determine an optimal learning rate by iteratively increasing λ until the objective function stops improving.
 - Just like discrete local search, gradient descent can get stuck in local optima in complex, non-convex functions. The same strategies of random restarts or more sophisticated exploration (analogous to GAs) are often employed.

- Many modern machine learning algorithms, particularly deep learning, rely heavily on gradient-based optimization, making the understanding of local search principles highly relevant.

Key Takeaways

- [] Local Beam Search improves upon single-node local search by maintaining a "beam" of K states, but can suffer from loss of diversity.
- [] Stochastic Local Beam Search addresses diversity loss by probabilistically selecting successors, biased towards fitter states.
- [] Genetic Algorithms are inspired by natural selection, using population, fitness functions, selection, crossing over, and mutation to explore search spaces.
- [] Crossing over in GAs enables non-local jumps, combining features from different parent solutions to potentially find novel, better states.
- [] Mutation in GAs introduces random changes, maintaining diversity and preventing premature convergence.
- [] The "champion" concept ensures the best-found solution is preserved across generations in GAs.
- [] GAs have many tunable parameters, and their performance can be sensitive to these choices.
- [] The core principles of local search, including the risk of local optima, apply to continuous optimization algorithms like Gradient Descent.
- [] Gradient Descent is the continuous space equivalent of local search, using gradients to determine the direction of improvement and a learning rate for step size.

Conclusion

The journey from simple hill climbing to sophisticated Genetic Algorithms and their continuous counterparts like Gradient Descent highlights the enduring power and adaptability of local search principles in AI. While systematic search guarantees optimality, its computational demands often render it impractical for real-world problems. Local search, in its various forms, offers a pragmatic alternative, providing good solutions efficiently, even if global optimality isn't always guaranteed. The evolution of these algorithms, from maintaining a single node to managing diverse populations and incorporating biological metaphors, showcases a continuous effort to balance exploitation of known good areas with exploration of the broader search space. Understanding these algorithms is crucial for anyone working in AI and machine learning, as their underlying intuitions form the bedrock of many state-of-

the-art optimization techniques, from training complex neural networks to solving combinatorial puzzles. The ongoing debate and resurgence of evolutionary algorithms further underscore their potential and the dynamic nature of research in this field.